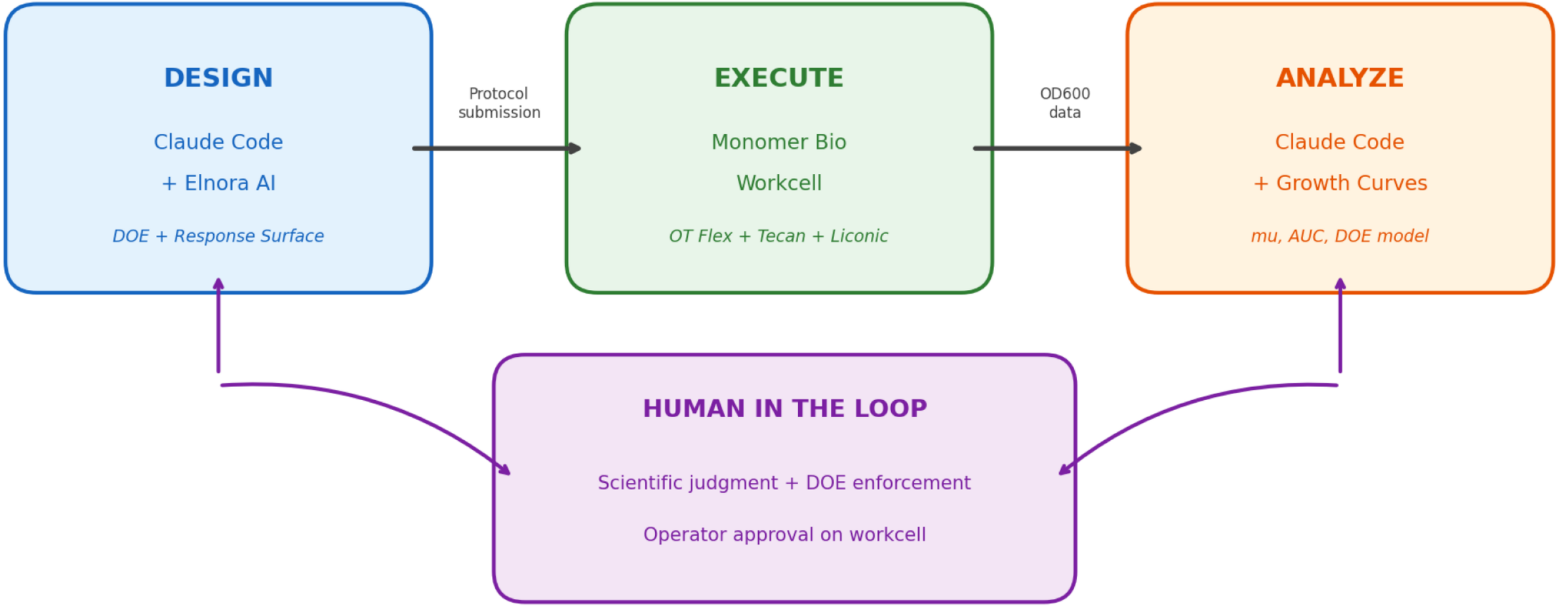
Autonomous Media Optimization for Vibrio natriegens

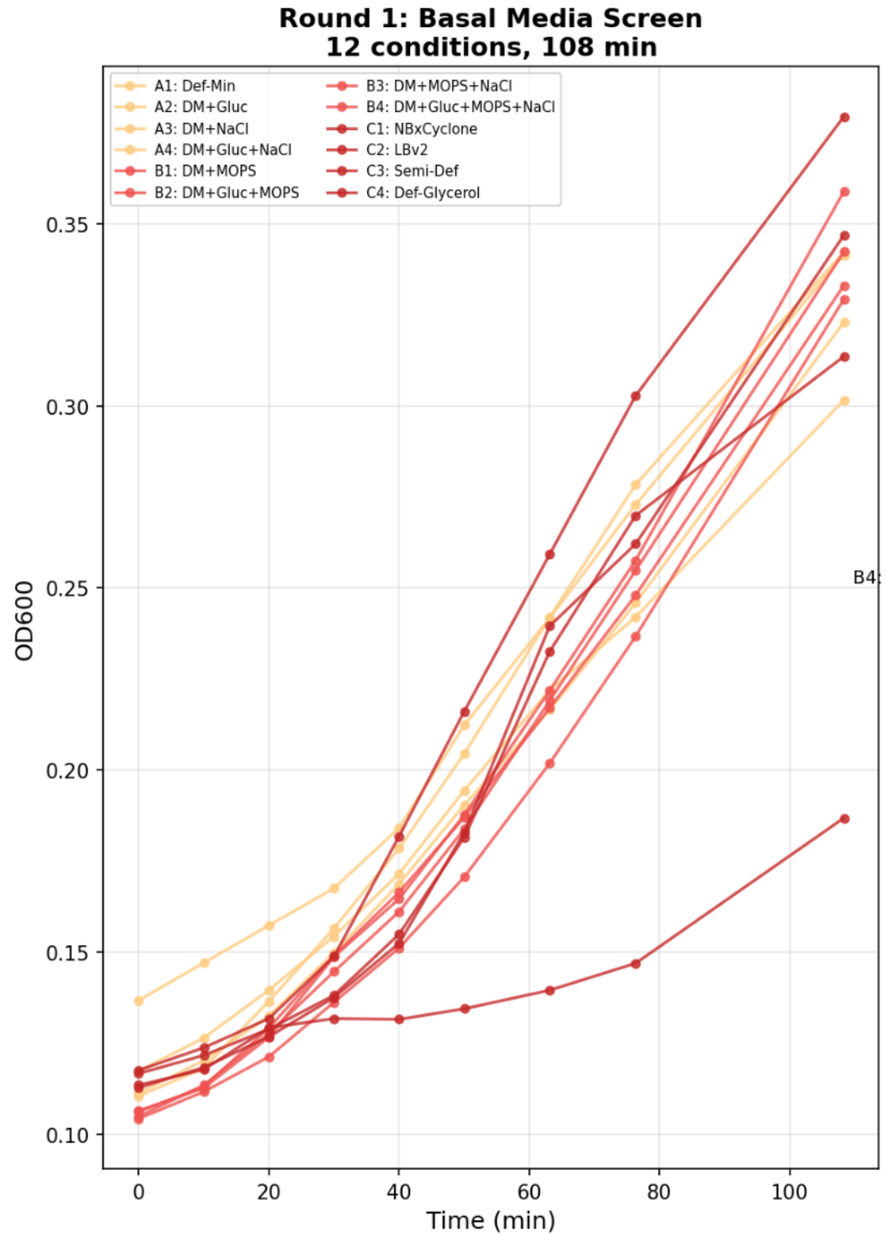
Team ViNatX — Monomer Bio / Elnora AI Science Hackathon

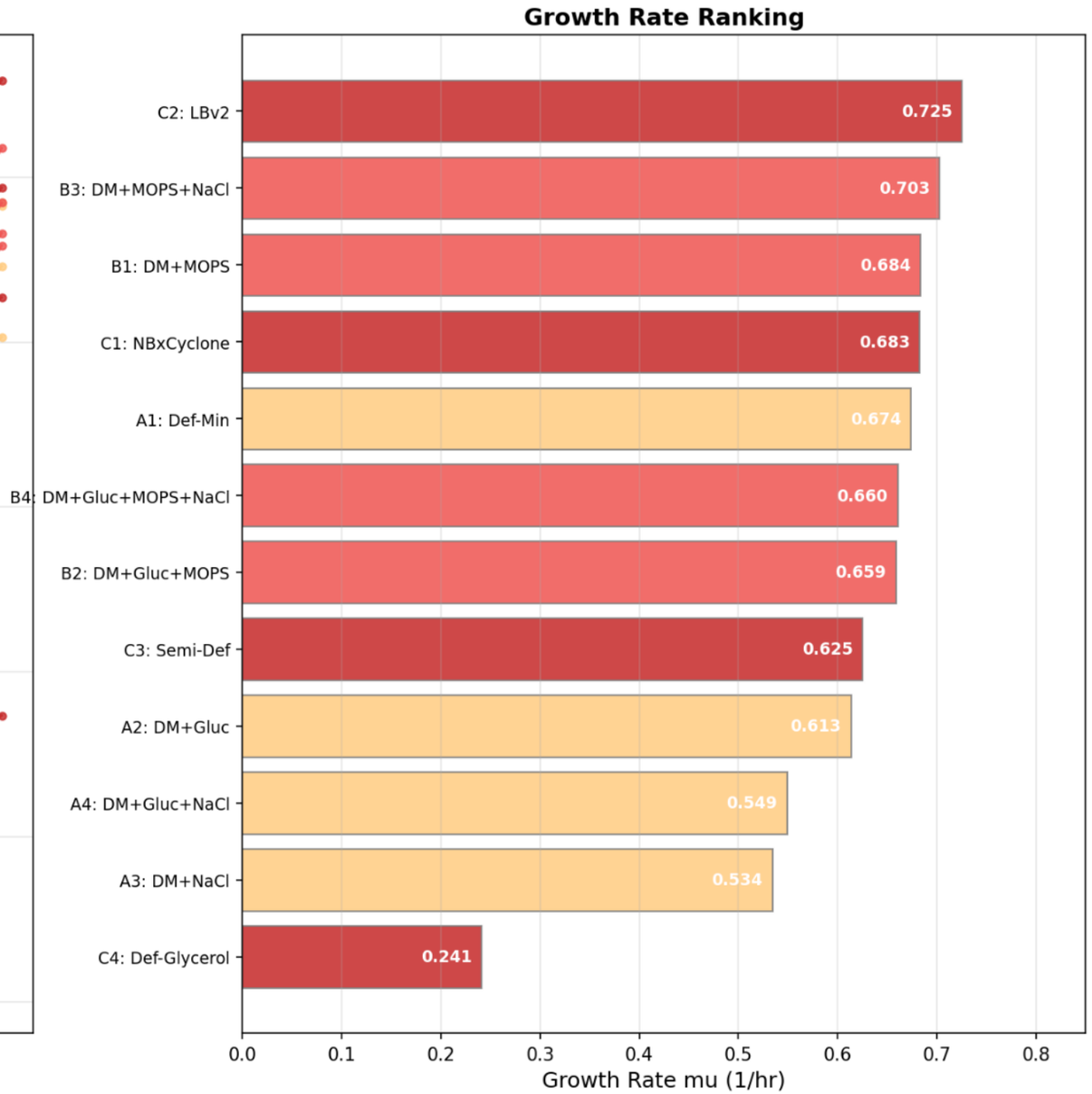


3 rounds completed: R1 (12 wells) → R2 (9 wells) → R3 (12 wells, running)

33 conditions tested across 3 iterative optimization cycles in ~14 hours

Phase 1: Which existing media formulations perform best?



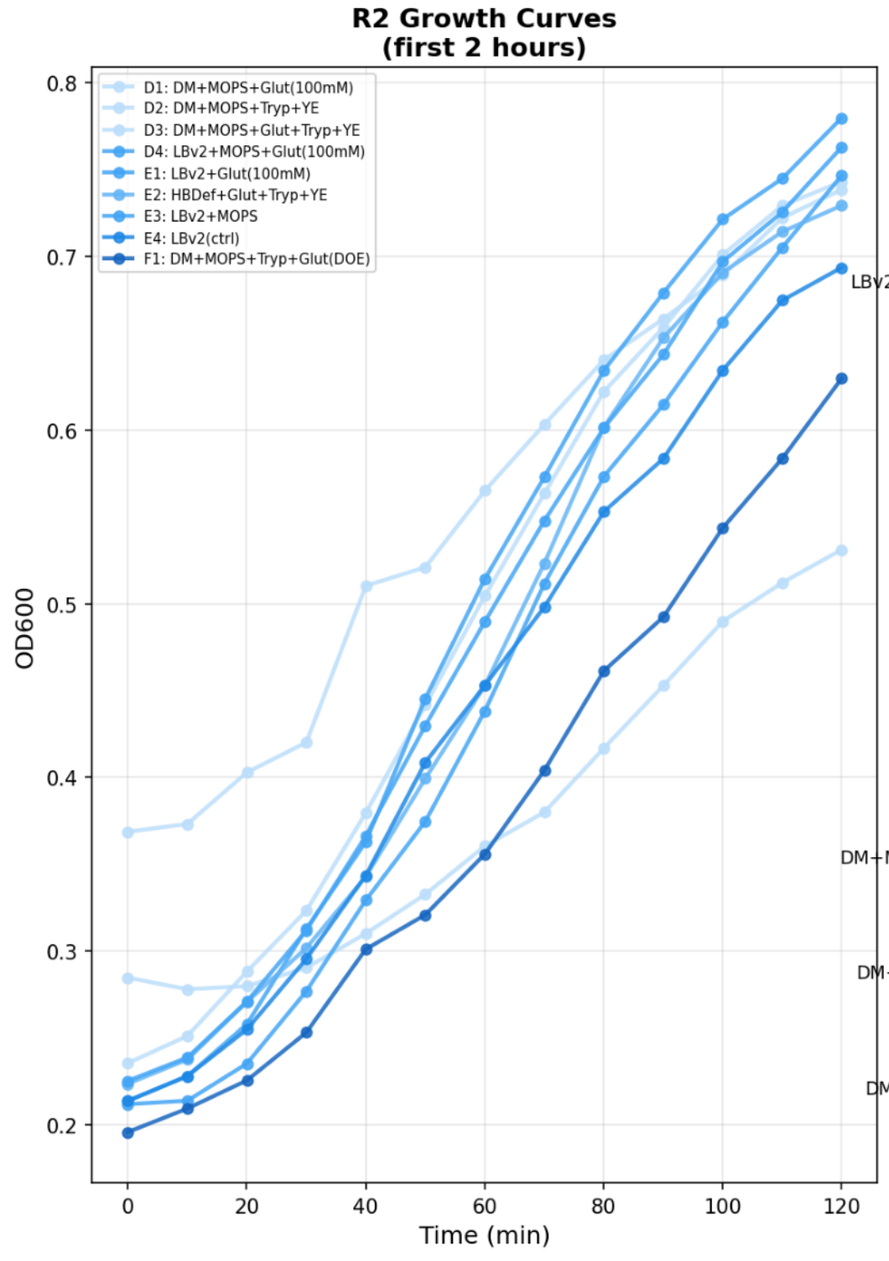


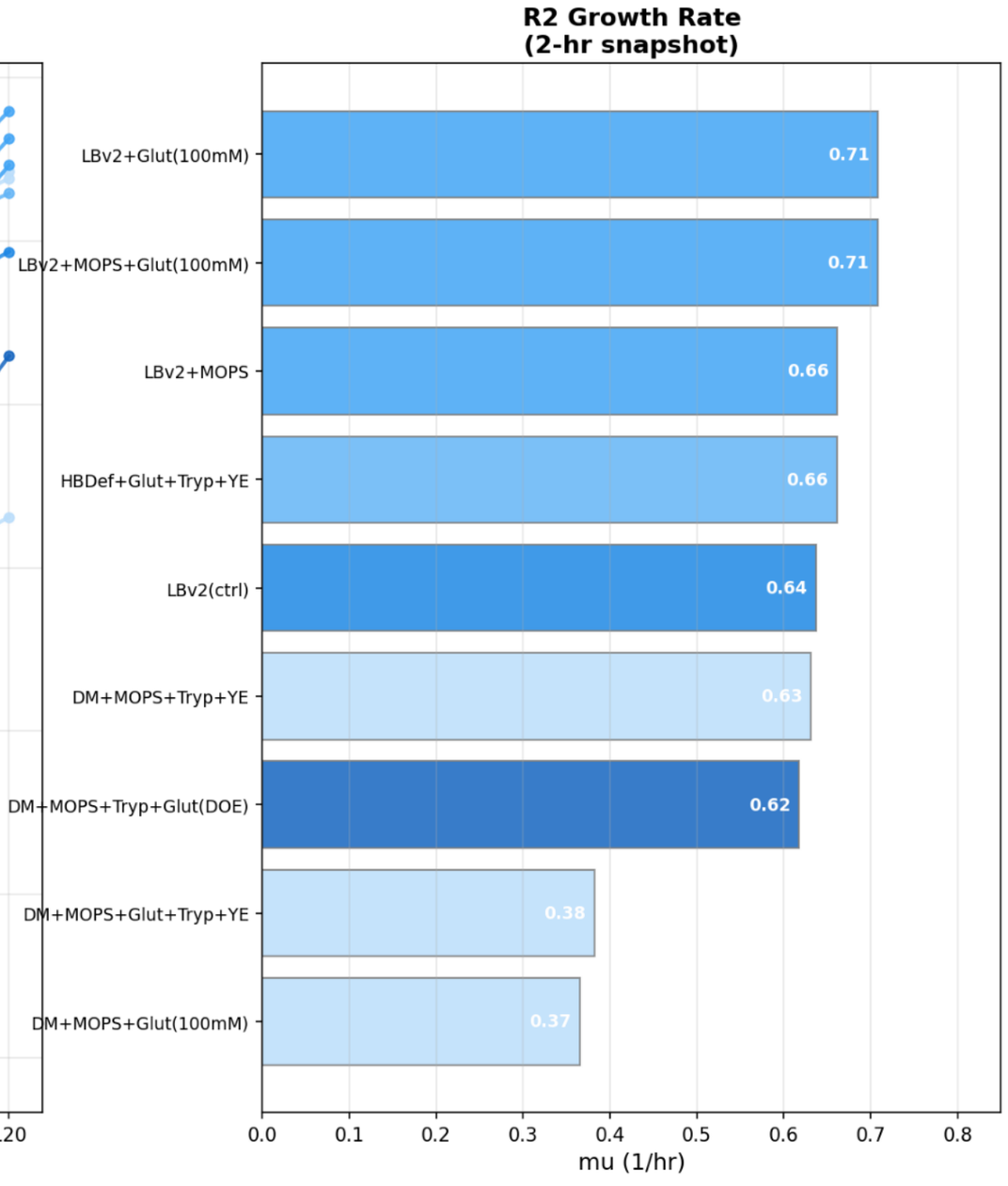
LBv2 >> Defined Media (pre-made peptides win)

MOPS buffer helps (+0.05 on DM base)

Glucose hurts NaCl hurts

Phase 2: Supplement optimization — what we learned at 2 hours + R3 design



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CONCLUSIONS FROM EARLY DATA:

1. LBv2 + Glut 100mM = best mu (0.709)

2. MOPS didn't help on LBv2
   (D4=0.708 vs E1=0.709)

3. 100mM Glut on Def-Min killed growth
   (D1=0.366, likely osmotic stress)

4. HBDef+supps (E2=0.661) didn't
   beat bare LBv2
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ROUND 3 DESIGN (12 wells, 40 transfers):

Glutamate dose-DOWN titration:
  G1: 10mM   G2: 25mM
  G3: 50mM   G4: 75mM

Glucose "supercharger" tests:
  H1: Glut 100mM + Glucose
  H4: Glucose only

Combo tests:
  H2: Glut 50mM + extra Tryp
  H3: Glut 50mM + extra YE

Controls:
  F2: Bare LBv2 (baseline drift)
  F3: Glut 100mM replicate
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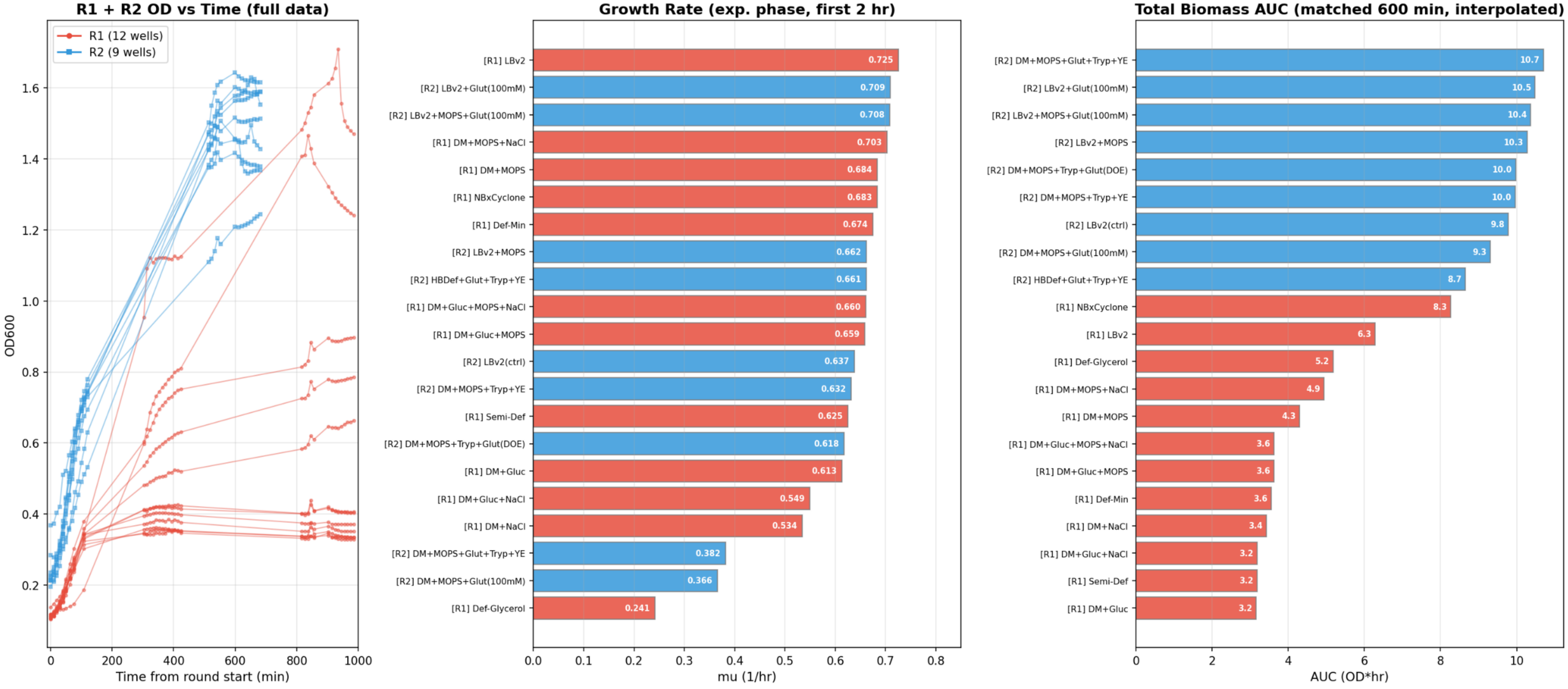
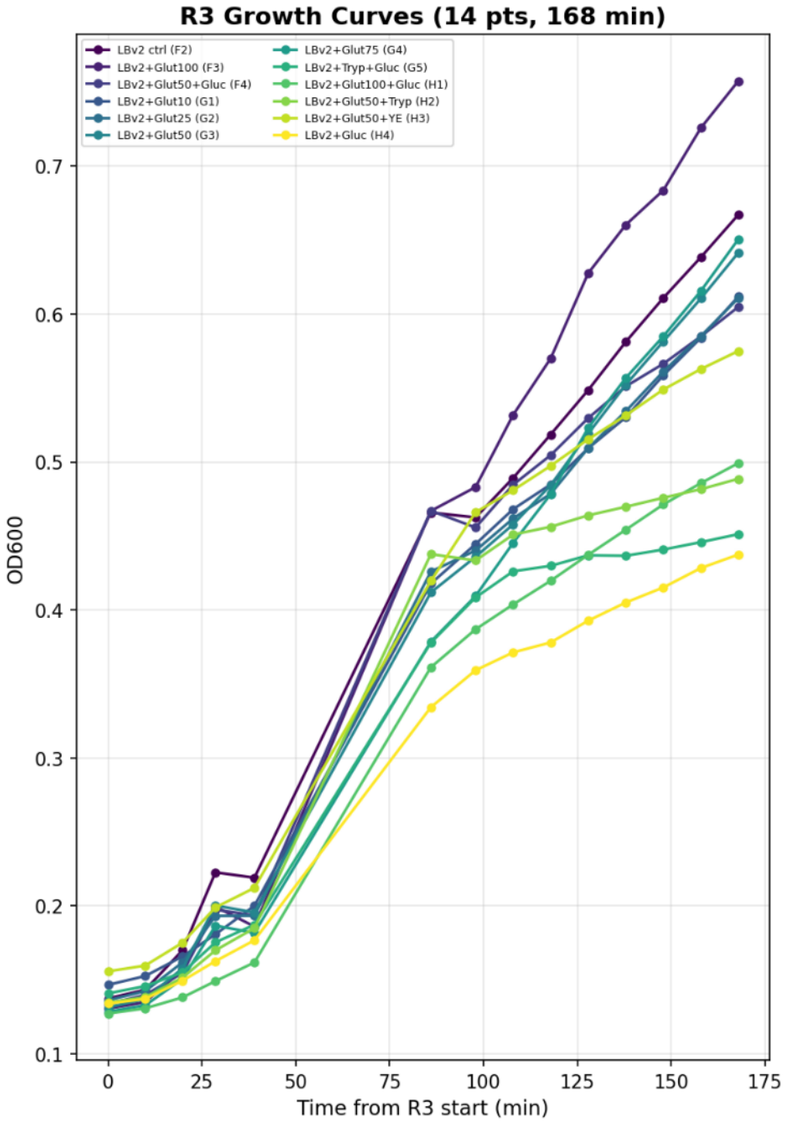
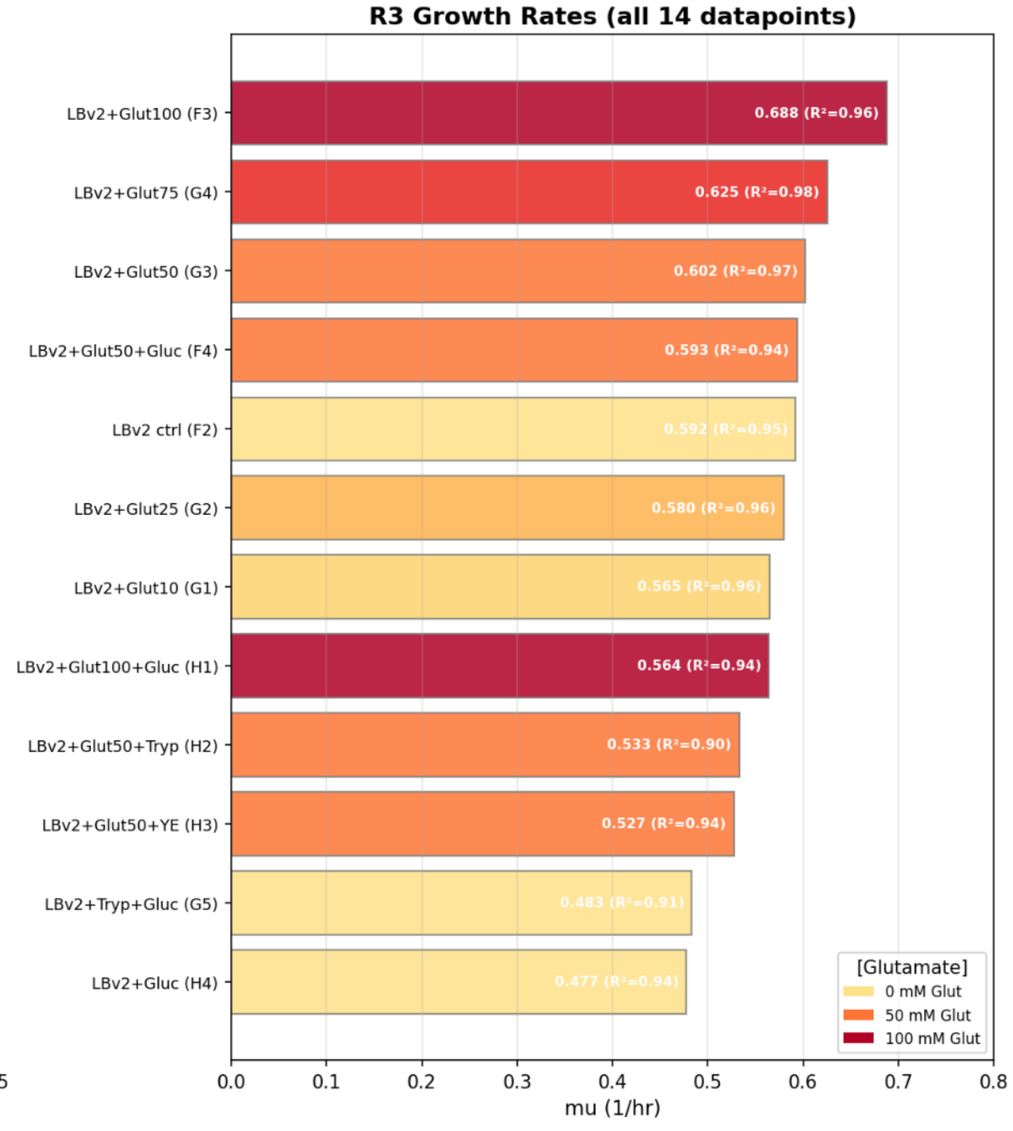
Phase 3: Extended data — R1 vs R2 comparison



Round 3 Results, Conclusions & Lessons Learned





KEY FINDINGS

1. Glutamate dose-response is clear
   100 > 75 > 50 > 25 > 10 mM
   Best: LBv2+Glut 100mM (mu=0.688)

2. Glutamate = best N-source
   Feeds directly into TCA cycle
   Outperforms tryptone and YE

3. Glucose HURTS growth rate
   Likely overflow metabolism

4. F3 replicates E1 (R2) well:
   mu 0.688 vs 0.709

WHAT WE LEARNED ABOUT AI + BIO

AI got us to hypotheses fast — initial DOE was wrong (glucose/NaCl on Def-Min), but rapid iteration let us pivot to glutamate + LBv2.

Integration was the bottleneck: MCP, Notion, Monomer took effort. Once connected, R2 and R3 moved much faster than R1.

AI made mistakes a grad student would also make, but lacked biases against "unusual" experiments.

16% IMPROVEMENT IN < 24 HOURS
LBv2+Glut 100mM (0.688) vs
bare LBv2 (0.592) — not plateaued
Next: titrate Glut UP + add MOPS